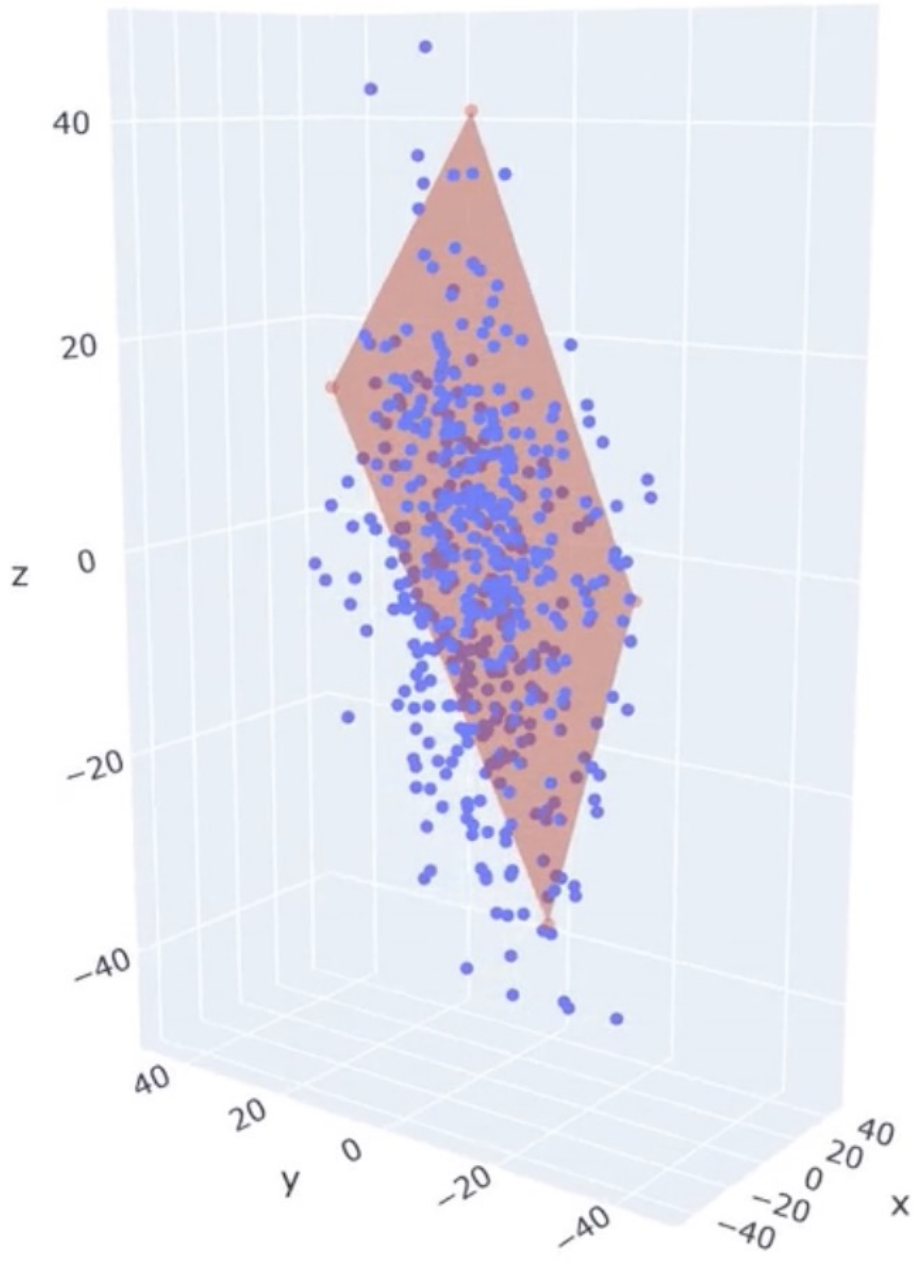
M-dimensional PCA

Archer Yang

McGill University

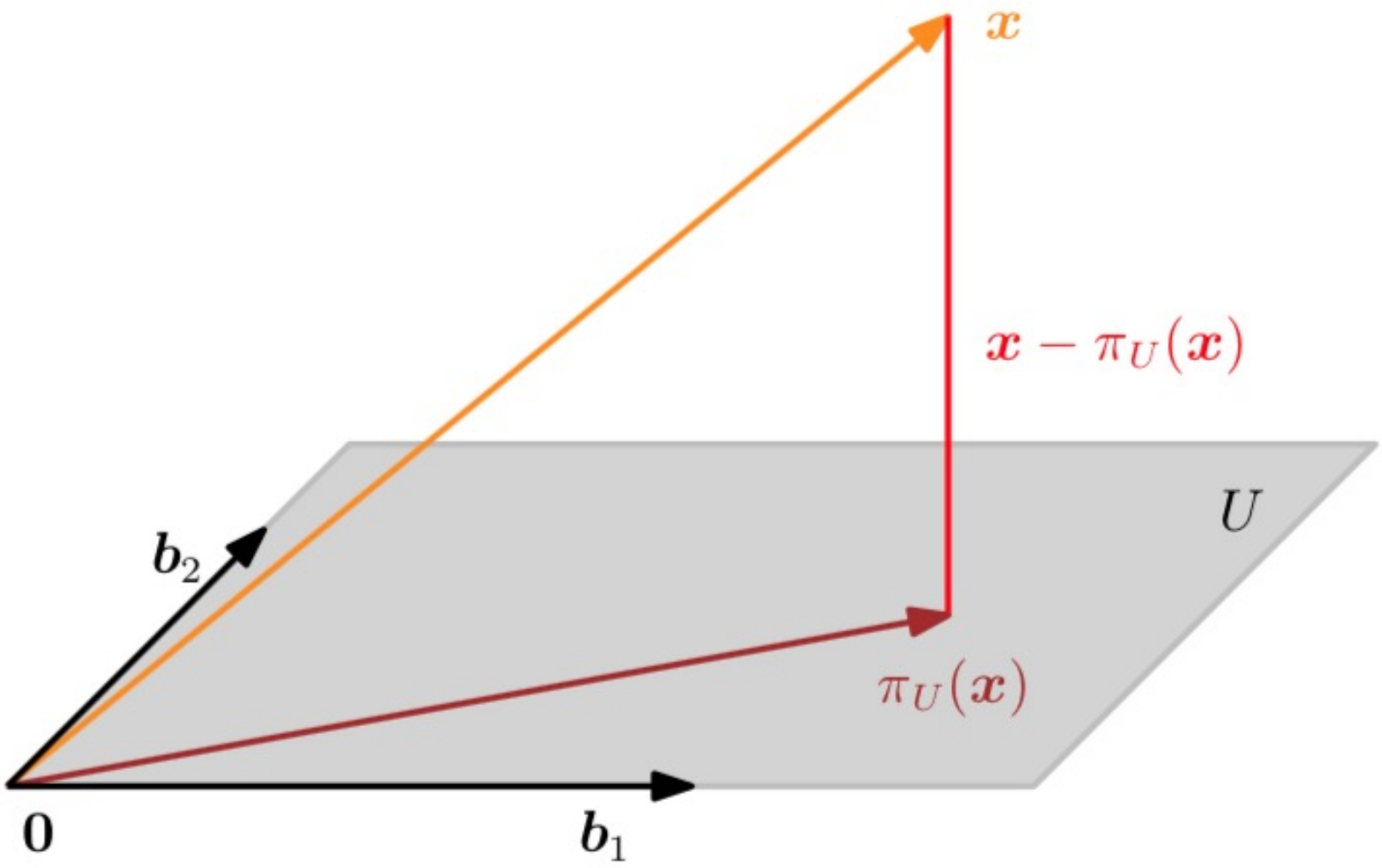
- **Reading assignment:** read chapter 4.1, 4.2 and 4.4 of Mathematics for Machine Learning book.

PCA: two-dimensional projection



Projection onto two-dimensional subspace

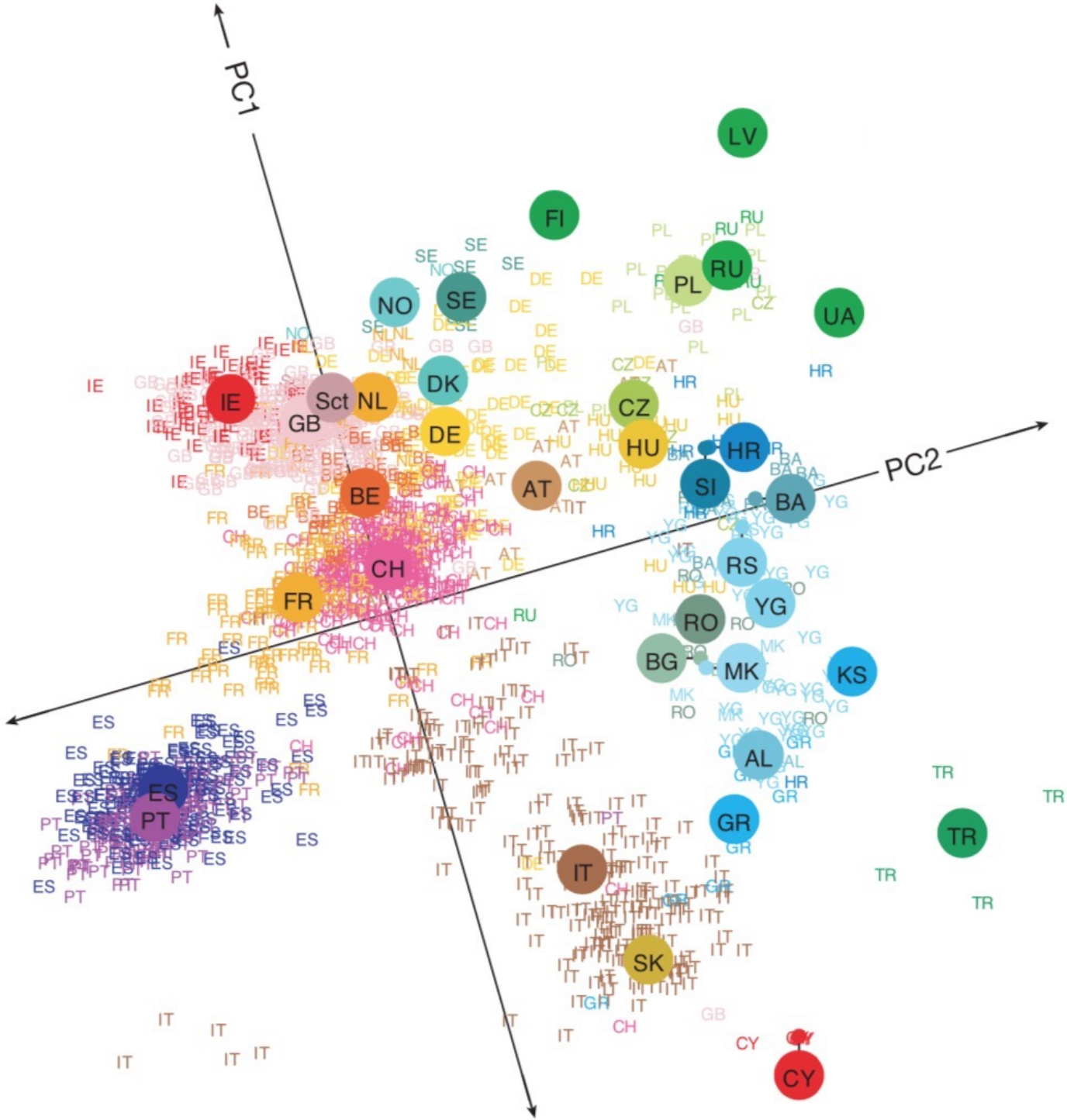
Now consider orthogonal projections onto two-dimensional subspaces $U \subseteq \mathbb{R}^n$.

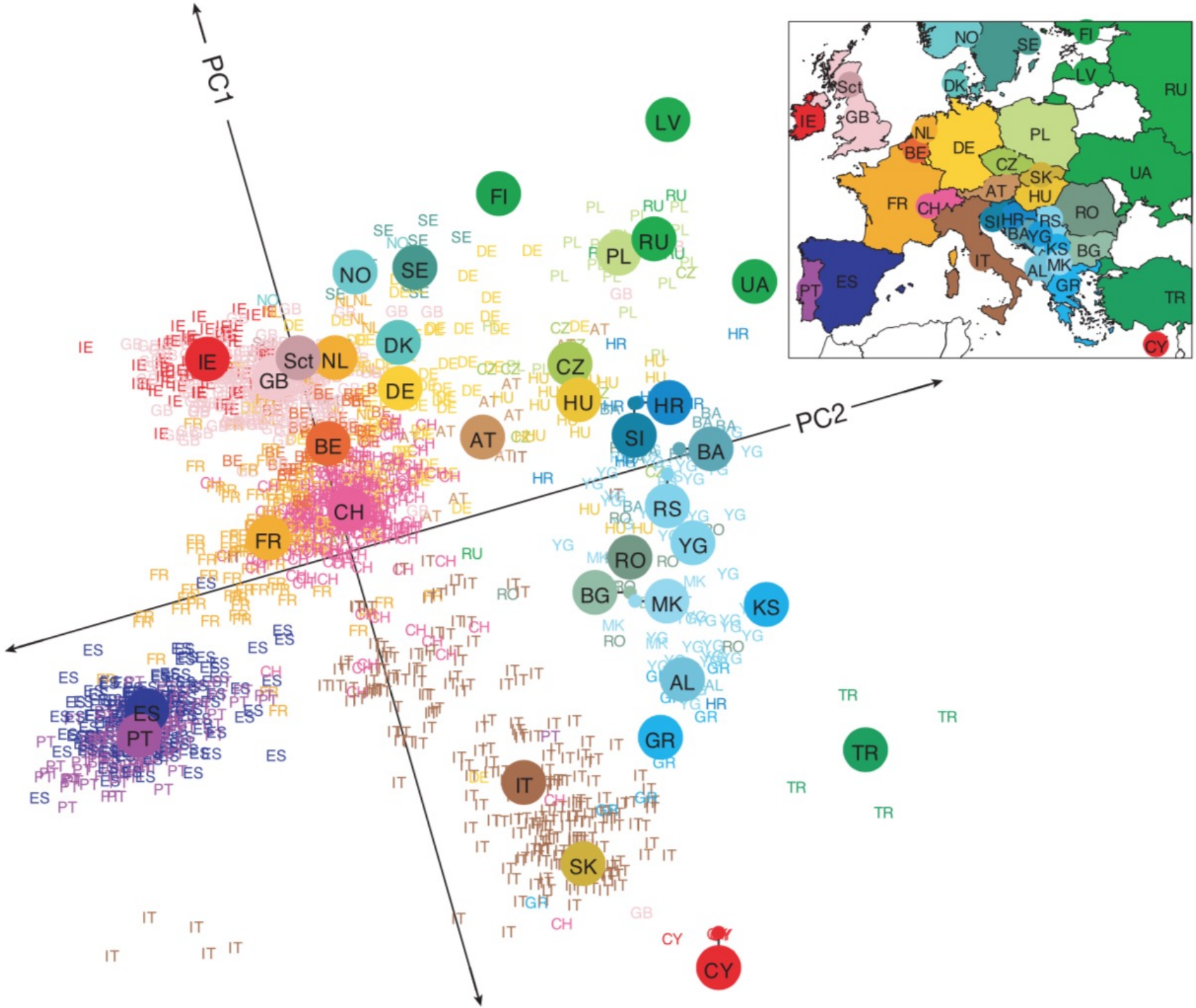


Application: population genetics

- **Data matrix:**
  - **Columns**: Genetic variation were genotyped at 500,568 loci using the SNP chip.
  - **Rows**: 3,192 European individuals were genotyped.
- Use the country of origin of each individual's grandparents to determine the geographic location.
- Exclude individuals with grand-parental ancestry from more than location

- Use PCA to produce a two-dimensional visual summary of the observed genetic variation.


PC1
PC2
LV
FI
RU
PL
UA
NO
SE
DK
IE
Sct
NL
GB
DE
CZ
HU
HR
SI
BA
AT
BE
CH
FR
RS
YG
RO
BG
MK
KS
AL
GR
TR
ES
PT
IT
SK
CY


PC1
PC2
LV
FI
RU
PL
NO
SE
UA
DK
IE
Sct
NL
GB
DE
CZ
HU
HR
SI
BA
AT
BE
CH
FR
RS
YG
RO
BG
MK
KS
AL
ES
PT
GR
TR
IT
SK
CY

PCA: Two-dimensional Case

Orthogonal projection

- We need to find:
  - first principal component $\boldsymbol{b}_1$
  - second principal component $\boldsymbol{b}_2$ – another unit vector orthogonal to $\boldsymbol{b}_1$.
- $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ form the orthonomal basis of a two-dimensional subspace U

$$\boldsymbol{b}_1^\top \boldsymbol{b}_2 = 0 \qquad ||\boldsymbol{b}_1|| = \boldsymbol{b}_1^\top \boldsymbol{b}_1 = 1 \qquad ||\boldsymbol{b}_2|| = \boldsymbol{b}_2^\top \boldsymbol{b}_2 = 1$$

Orthogonal projection

- The coordinate of the orthogonal projection for any data point $\boldsymbol{x}_n$ onto U is

$$\boldsymbol{z}_n = \begin{bmatrix} z_{1n} \\ z_{2n} \end{bmatrix} = \begin{bmatrix} \boldsymbol{b}_1^\top \boldsymbol{x}_n \\ \boldsymbol{b}_2^\top \boldsymbol{x}_n \end{bmatrix} = \boldsymbol{B}^\top \boldsymbol{x}_n \in \mathbb{R}^2$$

  where

$$\boldsymbol{B} = (\boldsymbol{b}_1, \boldsymbol{b}_2) \in \mathbb{R}^{D \times 2}$$

- The 2D vector $\boldsymbol{z}_n$ represents the coordinate of the projection of $\boldsymbol{x}_n$:
  - z_{1n} is the coordinate along direction $\boldsymbol{b}_1$
  - z_{2n} is along $\boldsymbol{b}_2$

Orthogonal projection

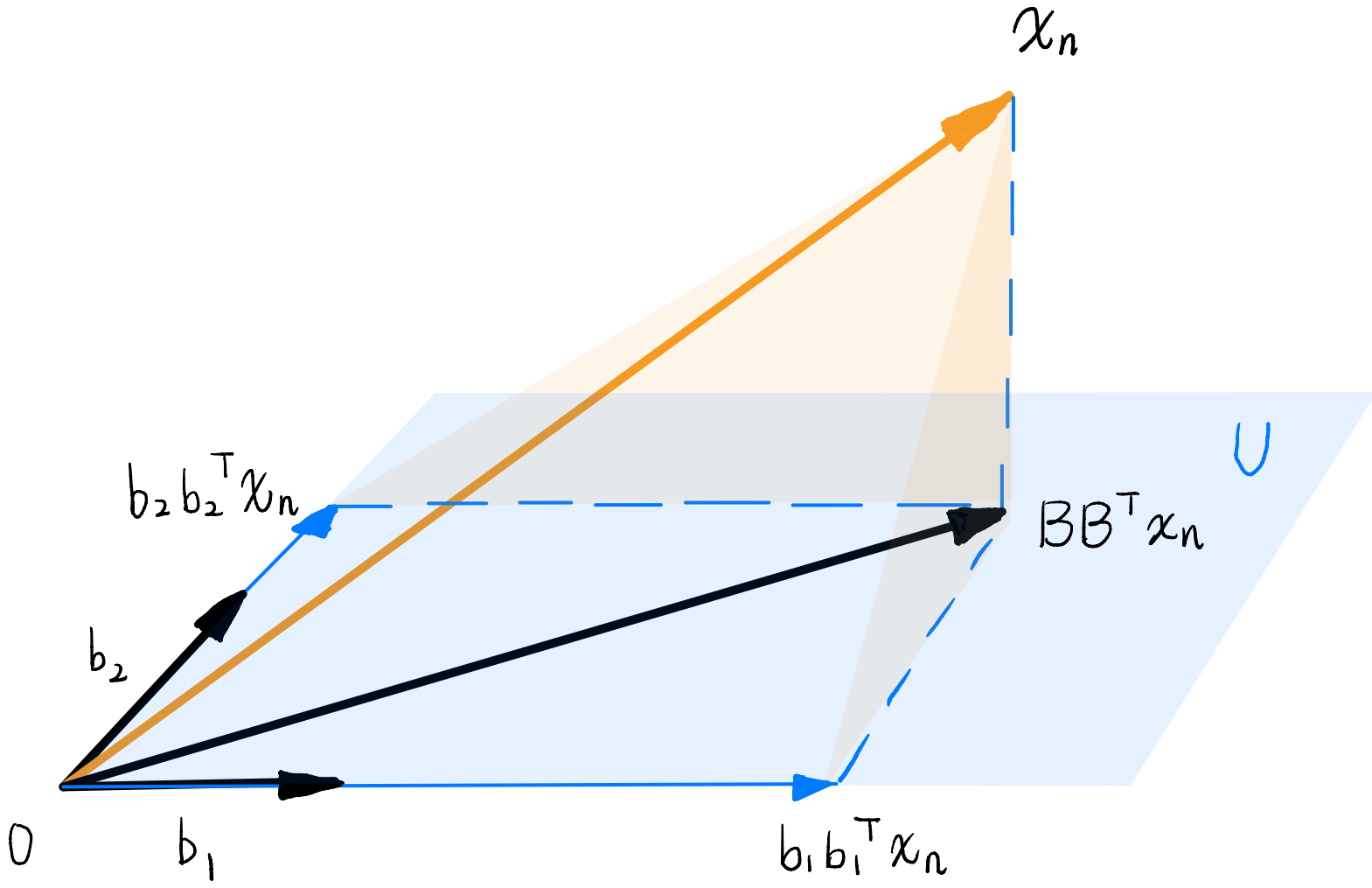
- The **projection** $\tilde{\boldsymbol{x}}_n$ of $\boldsymbol{x}_n$ is

$$\tilde{\boldsymbol{x}}_n = \boldsymbol{B}\boldsymbol{z}_n = \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x}_n$$

- Given N data points, $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N$, find $\boldsymbol{B}$ that minimizes the reconstruction error

$$\min_{\boldsymbol{B}} J_2 \equiv \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n\|^2 = \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x}_n\|^2$$

$$\text{subject to } \boldsymbol{B}^\top \boldsymbol{B} = \begin{bmatrix} \boldsymbol{b}_1^\top \boldsymbol{b}_1 & \boldsymbol{b}_1^\top \boldsymbol{b}_2 \\ \boldsymbol{b}_2^\top \boldsymbol{b}_1 & \boldsymbol{b}_2^\top \boldsymbol{b}_2 \end{bmatrix} = \boldsymbol{I}_{2\times 2} \tag{1}$$


x_n
$b_2 b_2^T x_n$
$BB^T x_n$
U
b_2
0
b_1
$b_1 b_1^T x_n$

Let's find optimal b_1 and b_2

- (Homework) One can show that

$$\tilde{\boldsymbol{x}}_n = \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x}_n = \boldsymbol{b}_1\boldsymbol{b}_1^\top \boldsymbol{x}_n + \boldsymbol{b}_2\boldsymbol{b}_2^\top \boldsymbol{x}_n$$

- Plug it into Eq. (1), the optimization problem (1) reduces to

$$\min_{\boldsymbol{b}_2} J_2 \equiv \frac{1}{N}\sum_{n=1}^{N} \| \underbrace{\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top \boldsymbol{x}_n}_{\text{residual}} - \boldsymbol{b}_2\boldsymbol{b}_2^\top \boldsymbol{x}_n \|^2$$

$$\text{subject to } \boldsymbol{b}_1^\top \boldsymbol{b}_2 = 0 \qquad \boldsymbol{b}_1^\top \boldsymbol{b}_1 = 1 \qquad \boldsymbol{b}_2^\top \boldsymbol{b}_2 = 1$$

- Observe that

$$
\begin{aligned}
J_2 &\equiv \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n - \boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n\|^2 \\
&= \frac{1}{N}\sum_{n=1}^{N} (\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n - \boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n)^\top(\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n - \boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n) \\
&= \frac{1}{N}\sum_{n=1}^{N} \Big(\boldsymbol{x}_n^\top\boldsymbol{x}_n - \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n - \boldsymbol{x}_n^\top\boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n \\
&\qquad - \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n + \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n + \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n \\
&\qquad - \boldsymbol{x}_n^\top\boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n + \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n + \boldsymbol{x}_n^\top\boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n\Big) \\
&= \frac{1}{N}\sum_{n=1}^{N} \Big(\boldsymbol{x}_n^\top\boldsymbol{x}_n - \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n - \boldsymbol{x}_n^\top\boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n\Big) \\
&= \text{const} - \boldsymbol{b}_1^\top\left(\frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}_n\boldsymbol{x}_n^\top\right)\boldsymbol{b}_1 - \boldsymbol{b}_2^\top\left(\frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}_n\boldsymbol{x}_n^\top\right)\boldsymbol{b}_2 \\
&= \text{const} - \boldsymbol{b}_1^\top\boldsymbol{S}\boldsymbol{b}_1 - \boldsymbol{b}_2^\top\boldsymbol{S}\boldsymbol{b}_2
\end{aligned}
$$

Equivalent formulation

- Therefore minimizing the reconstruction error is equivalent to maximizing the variance of the projection

$$\min_{\boldsymbol{b}_2} \frac{1}{N} \sum_{n=1}^{N} ||\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top \boldsymbol{x}_n - \boldsymbol{b}_2\boldsymbol{b}_2^\top \boldsymbol{x}_n||^2 \quad \Longleftrightarrow \quad \max_{\boldsymbol{b}_2} \boldsymbol{b}_1^\top \boldsymbol{S}\boldsymbol{b}_1 + \boldsymbol{b}_2^\top \boldsymbol{S}\boldsymbol{b}_2$$

$$\text{subject to } \boldsymbol{b}_1^\top \boldsymbol{b}_2 = 0 \qquad \text{subject to } \boldsymbol{b}_1^\top \boldsymbol{b}_2 = 0$$

$$\boldsymbol{b}_1^\top \boldsymbol{b}_1 = 1 \quad \boldsymbol{b}_2^\top \boldsymbol{b}_2 = 1 \qquad \boldsymbol{b}_1^\top \boldsymbol{b}_1 = 1 \quad \boldsymbol{b}_2^\top \boldsymbol{b}_2 = 1$$

(2)

To obtain optimal b_1

- Let's solve for b_1 first, we can ignore the grey part

$$\max_{b_2} b_1^\top S b_1 + b_2^\top S b_2 \quad \text{subject to } b_1^\top b_2 = 0 \quad b_1^\top b_1 = 1 \quad b_2^\top b_2 = 1$$

$$\iff \max_{b_2} b_1^\top S b_1 \quad \text{subject to } b_1^\top b_1 = 1$$

- b_1 is the **first eigenvector** of S – called the 1st **principal component (PC)**.

To obtain optimal b_2

- After the optimal b_1 is obtained, let's solve for b_2

$$\max_{b_2} b_1^\top S b_1 + b_2^\top S b_2$$

$$\text{subject to } b_1^\top b_2 = 0$$

$$b_1^\top b_1 = 1 \qquad b_2^\top b_2 = 1$$

To obtain optimal b_2

- To solve (2), introduce the **Lagrangian** based on contraints $\boldsymbol{b}_1^\top \boldsymbol{b}_2 = 0$ and $\boldsymbol{b}_2^\top \boldsymbol{b}_2 = 1$

$$\mathcal{L}(\boldsymbol{b}_2, \lambda_2, \lambda_{12}) = \boldsymbol{b}_2^\top \boldsymbol{S} \boldsymbol{b}_2 + \lambda_2(1 - \boldsymbol{b}_2^\top \boldsymbol{b}_2) + \lambda_{12}(0 - \boldsymbol{b}_1^\top \boldsymbol{b}_2)$$

- Compute the partial derivatives of $\mathcal{L}$ and set them to zero

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}_2} = 2\boldsymbol{b}_2^\top \boldsymbol{S} - 2\lambda_2 \boldsymbol{b}_2^\top - \lambda_{12} \boldsymbol{b}_1^\top = 0 \tag{3}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_2} = 1 - \boldsymbol{b}_2^\top \boldsymbol{b}_2 = 0 \qquad \frac{\partial \mathcal{L}}{\partial \lambda_{12}} = -\boldsymbol{b}_1^\top \boldsymbol{b}_2 = 0$$

To obtain optimal b_2

- Multiplying $\boldsymbol{b}_1$ on both side of Eq. (3) yields

$$2\boldsymbol{b}_2^\top \boldsymbol{S}\boldsymbol{b}_1 - 2\lambda_2 \boldsymbol{b}_2^\top \boldsymbol{b}_1 - \lambda_{12}\boldsymbol{b}_1^\top \boldsymbol{b}_1 = 0$$

$$2\boldsymbol{b}_2^\top \boldsymbol{S}\boldsymbol{b}_1 = \lambda_{12}$$

- On the other hand we know

$$\boldsymbol{S}\boldsymbol{b}_1 = \lambda_1 \boldsymbol{b}_1$$

Therefore

$$\lambda_{12} = 2\boldsymbol{b}_2^\top \lambda_1 \boldsymbol{b}_1 = 2\lambda_1 \boldsymbol{b}_2^\top \boldsymbol{b}_1 = 0 \Longrightarrow \lambda_{12} = 0$$

2nd Principal Component

- Plug it back into Eq. (3)

$$\boldsymbol{S}\boldsymbol{b}_2 = \lambda_2 \boldsymbol{b}_2$$

- To maximize the variance

$$\boldsymbol{b}_2^\top \boldsymbol{S}\boldsymbol{b}_2 = \boldsymbol{b}_2^\top \lambda_2 \boldsymbol{b}_2 = \lambda_2$$

  we choose the basis vector $\boldsymbol{b}_2$ associated with the **second largest eigenvalue** of the data covariance matrix $\boldsymbol{S}$.

- The eigenvector $\boldsymbol{b}_2$ is called the 2nd **principal component (PC)**.

PCA results

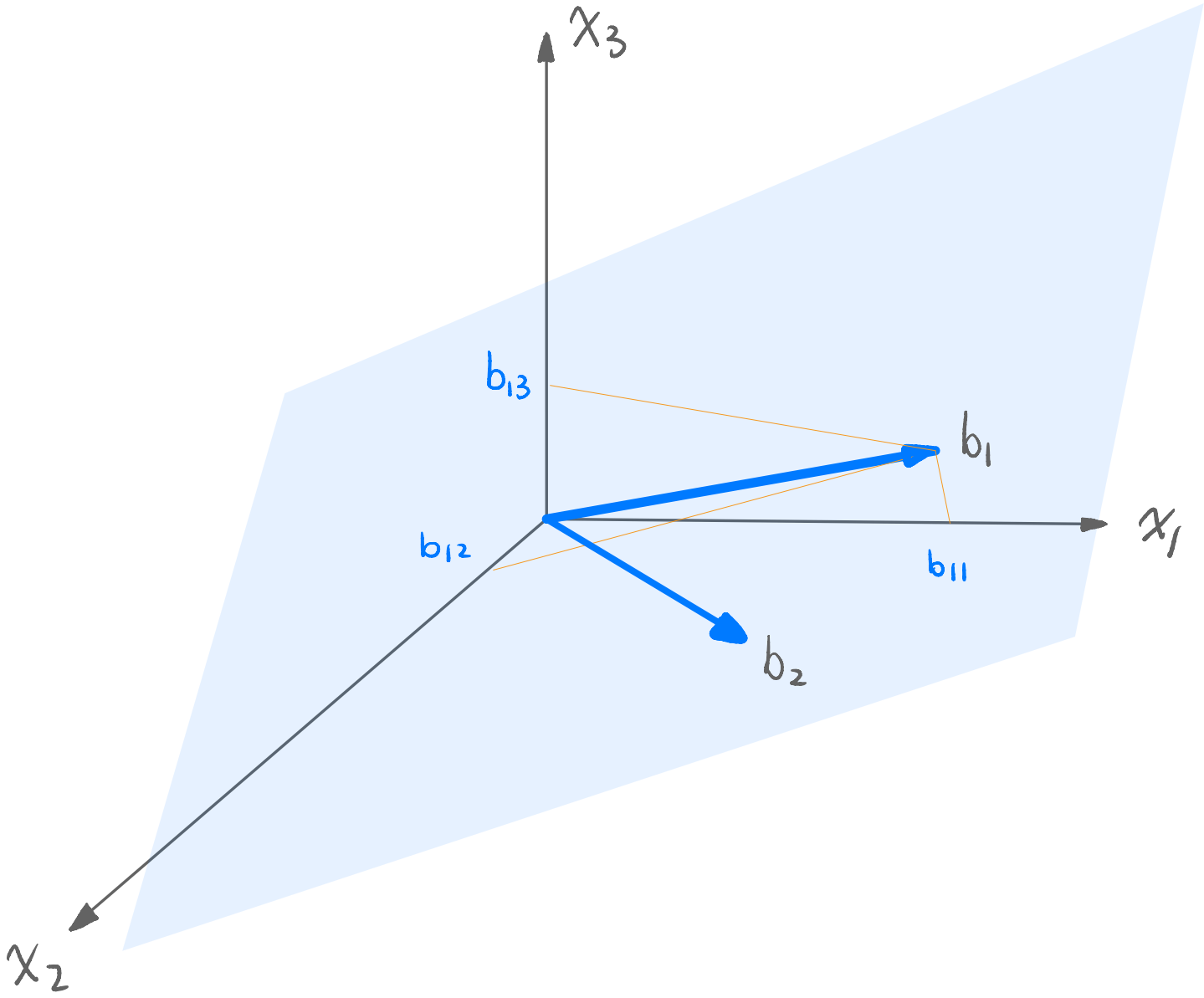
- The optimal $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ are **1st and 2nd principal component (PC)**.

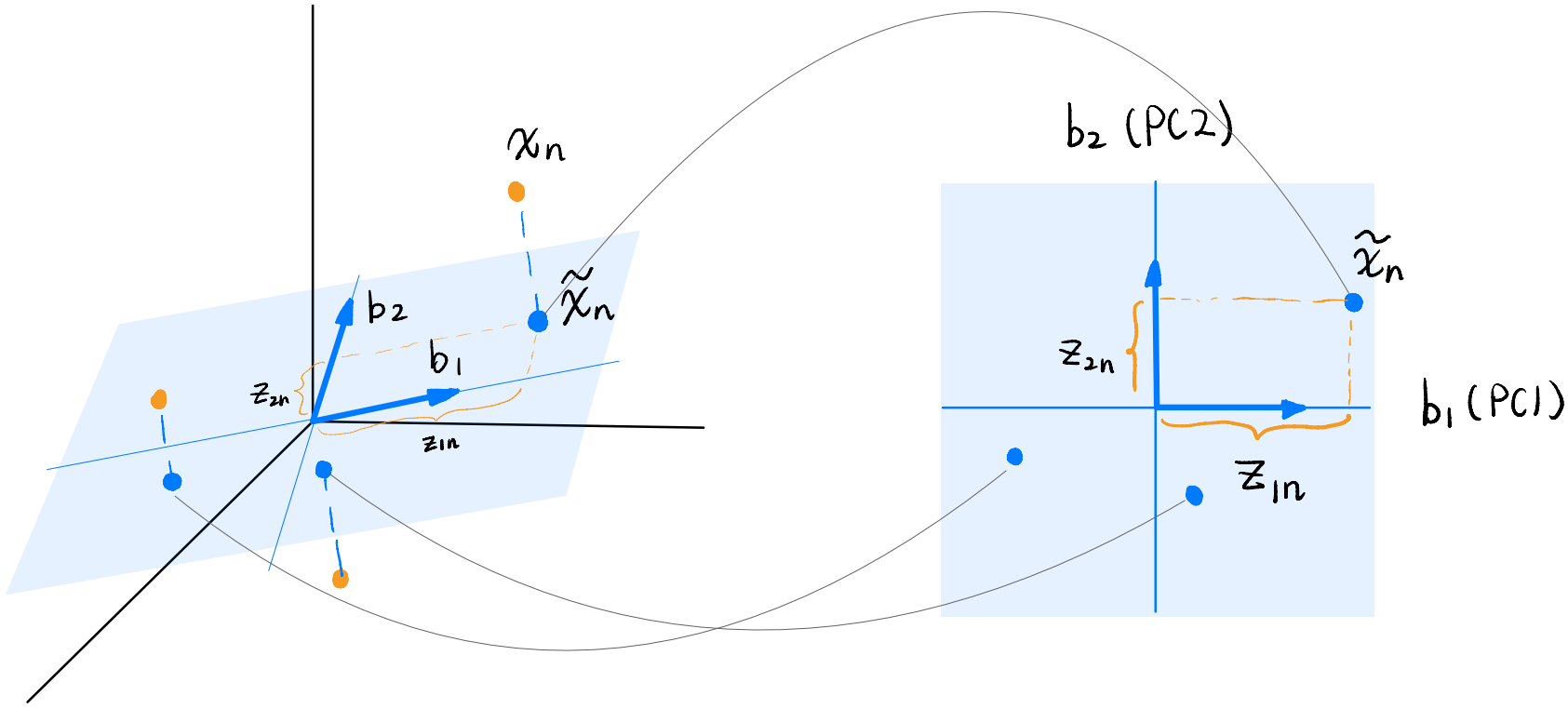
$$\boldsymbol{B} = (\boldsymbol{b}_1, \boldsymbol{b}_2) \in \mathbb{R}^{D \times 2}$$

- z_{1n} and z_{2n} are **PC1 and PC2 scores**

$$\boldsymbol{z}_n = (z_{1n}, z_{2n})^\top = \boldsymbol{B}^\top \boldsymbol{x}_n \in \mathbb{R}^2$$

- z_{1n} is the coordinate of the projection along direction $\boldsymbol{b}_1$, z_{2n} is along $\boldsymbol{b}_2$


x_3
b_{13}
b_1
x_1
b_{12}
b_{11}
b_2
x_2


x_n
$\tilde{x}_n$
b_2
b_1
z_{2n}
z_{1n}
b_2 (PC2)
$\tilde{x}_n$
z_{2n}
b_1 (PC1)
z_{1n}

- (Homework) When $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ are the **first** and **second** principal components, respectively, their corresponding scores $z_{11}, \cdots, z_{1N}$ and $z_{21}, \cdots, z_{2N}$ have **zero** sample covariance, i.e. PC1 scores appears **uncorrelated** with PC2 scores.

$$\frac{1}{N}\sum_{n=1}^{N}(z_{1n} - \bar{z}_1)(z_{2n} - \bar{z}_2) = \frac{1}{N}\sum_{n=1}^{N} z_{1n}z_{2n} = 0$$

- (Homework) When $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ are the **first** and **second** principal components, respectively, the re-construction error is equal to

$$J_2 \equiv \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top \boldsymbol{x}_n - \boldsymbol{b}_2\boldsymbol{b}_2^\top \boldsymbol{x}_n\|^2$$

$$= \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n^\top \boldsymbol{x}_n - \lambda_1 - \lambda_2$$

Another perspective of reconstruction error

- (Homework) Define the residual $\boldsymbol{r}_n$ as the difference between $\boldsymbol{x}_n$ and approximation $\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n$

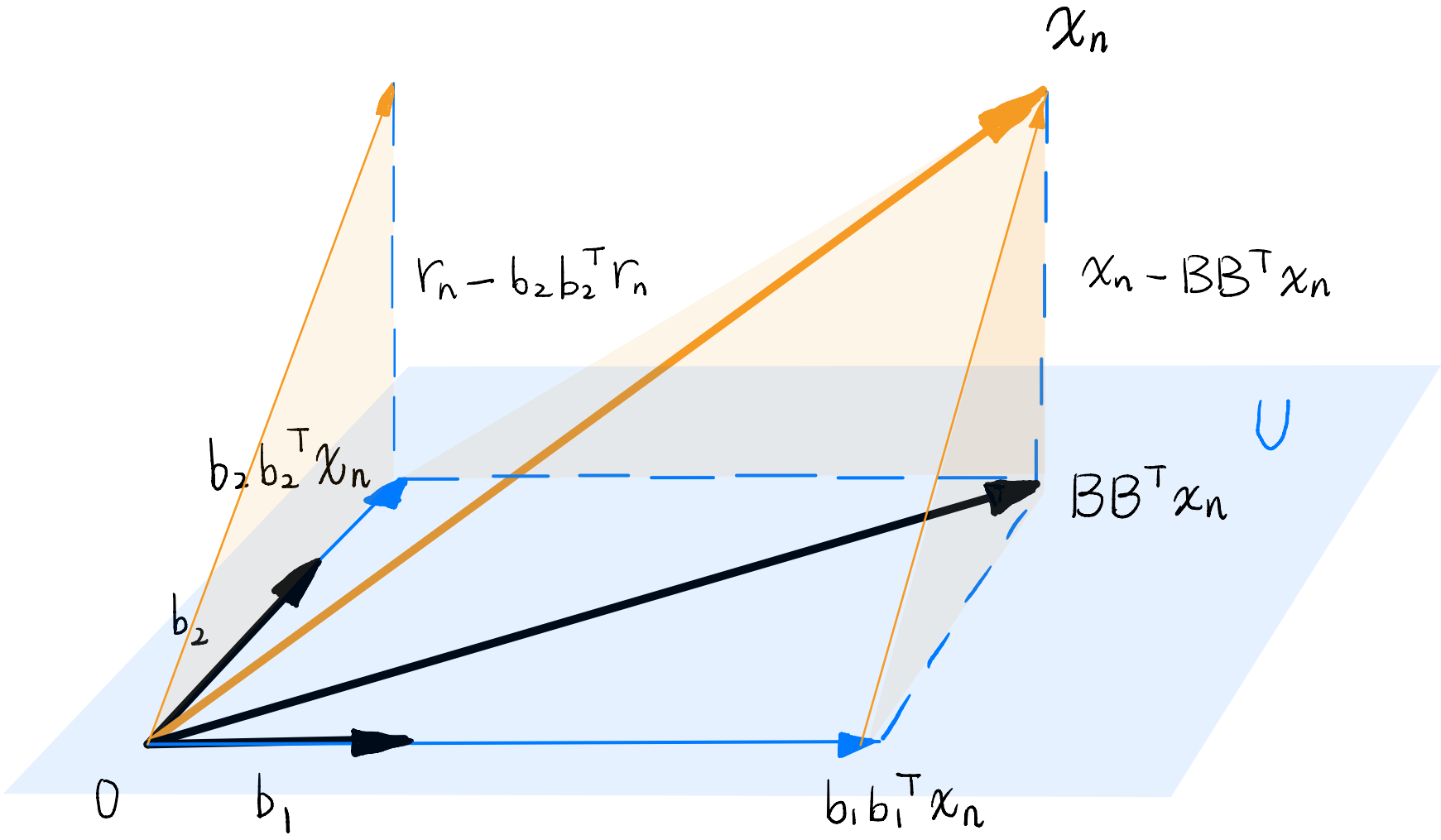
$$\boldsymbol{r}_n = \boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n.$$

Show that

$$J_2 \equiv \frac{1}{N}\sum_{n=1}^{N} \| \underbrace{\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n}_{\boldsymbol{r}_n\text{:residual}} - \boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{x}_n \|^2 = \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{r}_n - \boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{r}_n\|^2$$

Another perspective of reconstruction error

- This implies that J_2 can be viewed as the **reconstruction error between** the "residual" $\boldsymbol{r}_n$ (obtained by projecting $\boldsymbol{x}_n$ onto $\boldsymbol{b}_1$) and the projected residual $\boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{r}_n$ (obtained by projecting $\boldsymbol{r}_n$ onto $\boldsymbol{b}_2$).
- i.e. $\boldsymbol{r}_n - \boldsymbol{b}_2\boldsymbol{b}_2^\top\boldsymbol{r}_n$ is **"the residual of the residual"**.
- It also implies: if we treat the residuals $\boldsymbol{r}_n$ for $n = 1, \cdots, N$ as new data, and ask for their "1st PC". This will be the direction of largest variance which is **perpendicular** to the 1st PC of the original data. In other words, it will be the 2nd PC of the original data.


x_n
$r_n - b_2 b_2^T r_n$
$x_n - BB^T x_n$
$b_2 b_2^T x_n$
$BB^T x_n$
U
b_2
0
b_1
$b_1 b_1^T x_n$

Another perspective of reconstruction error

- This suggests a recursive algorithm for finding all the principal components for the PCA in the general M-dimensional case.
- The m-th principal component is the leading component of the residuals after subtracting off the first $m - 1$ components.

PCA: M-dimensional Case

Projection onto M-dimensional subspace

- Given a dataset of D dimensions, i.e. $\boldsymbol{x}_n \in \mathbb{R}^D$ for $n = 1, \cdots, N$.
- We project data points $\boldsymbol{x}_n$ for $n = 1, \cdots, N$ to a M-dimensional subspace ($M \leq D$) spanned by basis $\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M$.
- Similar to the 2D case, **in a sequential manner**, we can obtain $\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M$ and corresponding eigenvalues $\lambda_1, \cdots, \lambda_M$.
- In practice, it is faster to use eigenvector-solvers to get all $\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M$ and $\lambda_1, \cdots, \lambda_M$ at once.

Diagonalization theorem and eigendecomposition (Thm. 4.20 MML)

A $D \times D$ square matrix $\boldsymbol{A}$ is diagonalizable **if and only if** $\boldsymbol{A}$ has D linearly independent eigenvectors. In this case,

$$\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}$$

with **invertible matrix** $\boldsymbol{V}$ of eigenvectors and **diagonal matrix** $\boldsymbol{\Lambda}$ of eigenvalues

$$\boldsymbol{V} = \begin{pmatrix} | & | & & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_D \\ | & | & & | \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_D \end{pmatrix}$$

where $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_D$ are linearly independent eigenvectors, and $\lambda_1, \lambda_2, \cdots, \lambda_D$ are the corresponding eigenvalues, in the same order.

Eigendecomposition

Eigendecomposition for a symmetric matrix (Thm. 4.21 MML)

A **symmetric** matrix $\boldsymbol{A} \in \mathbb{R}^{D\times D}$ can always be diagonalized, i.e.

$$\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\top} = \sum_{i=1}^{D} \lambda_i \boldsymbol{v}_i \boldsymbol{v}_i^{\top}$$

where $\boldsymbol{V}$ is an orthogonal matrix containing eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_D$ that forms **an orthonormal basis** of $\mathbb{R}^D$

$$\boldsymbol{V} = \begin{pmatrix} | & | & & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_D \\ | & | & & | \end{pmatrix}, \qquad \boldsymbol{V}^{\top}\boldsymbol{V} = \mathbf{I}_{D\times D}$$

implying that $\boldsymbol{V}^{-1} = \boldsymbol{V}^{\top}$; and $\boldsymbol{\Lambda}$ is a diagonal matrix whose diagonal entries are the eigenvalues of $\boldsymbol{A}$.

Eigendecomposition

- If the data are projected onto a M-dimensional subspace ($M \leq D$), PCA boils down to computing the M eigenvectors of the sample covariance matrix $\boldsymbol{S}$ that have the largest eigenvalues.
- (Thm. 4.14 MML) One can show that $\boldsymbol{S} = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n \boldsymbol{x}_n^\top = \frac{1}{N}\boldsymbol{X}^\top \boldsymbol{X}$ is always a **symmetric** matrix. Thus the eigendecomposition

$$\boldsymbol{S} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^\top$$

always exists.

- The principal components $\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M$ are the first M columns of orthogonal matrix $\boldsymbol{V}$.

$$\boldsymbol{B} = \begin{pmatrix} | & | & & | \\ \boldsymbol{b}_1 & \boldsymbol{b}_2 & \cdots & \boldsymbol{b}_M \\ | & | & & | \end{pmatrix} \in \mathbb{R}^{D \times M}$$

$$\| $$

$$\boldsymbol{V} = \begin{pmatrix} | & | & & | & | & & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_M & \boldsymbol{v}_{M+1} & \cdots & \boldsymbol{v}_D \\ | & | & & | & | & & | \end{pmatrix}$$

  - The corresponding eigenvalues $\lambda_1, \lambda_2, \cdots, \lambda_M$ are the largest M diagonal entries of $\boldsymbol{\Lambda}$.

Some Insights in M-dimensional Case

Reconstruction error and variance

- (Homework) Show that the reconstruction error in the M-dimensional project case is

$$J_M = \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x}_n\|^2 = \frac{1}{N}\sum_{n=1}^{N} \left\|\boldsymbol{x}_n - \sum_{j=1}^{M} \boldsymbol{b}_j \boldsymbol{b}_j^\top \boldsymbol{x}_n\right\|^2$$

$$= \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n^\top \boldsymbol{x}_n - \underbrace{\sum_{j=1}^{M} \boldsymbol{b}_j^\top \boldsymbol{S} \boldsymbol{b}_j}_{V_M}$$

- (Homework) Furthermore, show that if $\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M$ are the first M PCs and $\lambda_1, \cdots, \lambda_M$ are the corresponding eigenvalues, then

$$J_M = \underbrace{\frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n^\top \boldsymbol{x}_n}_{\text{Total var.}} - \underbrace{\sum_{j=1}^{M} \lambda_j}_{}$$

V_M : var. captured by first M PCs

Total variance

- (Homework) Show that the constant term $\frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n^\top \boldsymbol{x}_n$ represents the total variance in the data

$$\frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n^\top \boldsymbol{x}_n = \operatorname{tr}\left(\frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n \boldsymbol{x}_n^\top\right) = \operatorname{tr}\left(\frac{1}{N}\boldsymbol{X}^\top \boldsymbol{X}\right) = \operatorname{tr}(\boldsymbol{S}) \overset{(i)}{=} \sum_{j=1}^{D} \lambda_j$$

  Eq. (i) follows from Thm. 4.17 MML

- This implies that

$$J_M = \sum_{j=1}^{D} \lambda_j - \sum_{j=1}^{M} \lambda_j = \underbrace{\sum_{j=M+1}^{D} \lambda_j}_{\text{uncaptured var. by first } M \text{ PCs}}$$

Total variance

- **Fruit for thought:** what does it tell us if eigenvalues $\lambda_{M+1}, \cdots, \lambda_D$ are zero?

$$J_M = \sum_{j=1}^{D} \lambda_j - \sum_{j=1}^{M} \lambda_j = \underbrace{\sum_{j=M+1}^{D} \lambda_j}_{\text{uncaptured var. by first } M \text{ PCs}} = 0$$

- In such case, what would be the **intrinsic dimension** of the data?

Low-rank matrix perspective

- Sample covariance matrix $\boldsymbol{S} = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n \boldsymbol{x}_n^\top = \frac{1}{N}\boldsymbol{X}^\top \boldsymbol{X}$ is always **symmetic** and **positive semidefinite**. (Thm. 4.14 MML)
- This implies that $\lambda_1, \cdots, \lambda_D \geq 0$
- We have

$$\operatorname{rank}(\boldsymbol{S}) = \operatorname{rank}\left(\frac{1}{N}\boldsymbol{X}^\top \boldsymbol{X}\right) = \operatorname{rank}(\boldsymbol{X})$$

$$= \text{\# of nonzero eigenvalues of } \boldsymbol{S}$$

Low-rank matrix perspective

- Therefore if $\lambda_{M+1} = \lambda_{M+2} = \cdots = \lambda_D = 0$, it implies that $\mathrm{rank}(\boldsymbol{X}) = M$.
- e.g. If $M = 2$ and $\lambda_3 = \cdots = \lambda_D = 0$, we have $\mathrm{rank}(\boldsymbol{X}) = 2$.
- This implies that the intrinsic dimension of the data is two. i.e. we can use the linear combination of any two columns to represent any other $N - 2$ columns of $\boldsymbol{X}$ – high dependence between variables.